

HW1 - Stochastic Methods for Industrial Engineers

Submission: June 2, 2026

1 True or False?

For each of the following claims, determine if they are true or false. If you think they are true - prove them. If you think they are false - find a counterexample. Explain your answers clearly.

- a) (10 points) Let $X_1, X_2, \dots$ be a sequence of independent random variables, where for each n , X_n is uniformly distributed over the union of intervals

$$I_n = \bigcup_{k=0}^{n-1} \left[\frac{2k}{2n}, \frac{2k+1}{2n} \right].$$

Let $Z \sim U(0, 1)$ be a random variable independent of the sequence. It is always true that $X_n \xrightarrow{d} Z$.

- (a) **True.** I_n consists of n intervals of length $\frac{1}{2n}$, so its total length is $\frac{1}{2}$. Let F_n be the CDF of X_n . For $x \in [0, 1]$, the amount of I_n contained in $[0, x]$ differs from $\frac{x}{2}$ by at most the length of one interval, which is $\frac{1}{2n}$. Therefore

$$\left| |I_n \cap [0, x]| - \frac{x}{2} \right| \leq \frac{1}{2n}.$$

Since X_n is uniform on a set of length $\frac{1}{2}$, then

$$F_n(x) = \frac{|I_n \cap [0, x]|}{1/2}.$$

Thus

$$|F_n(x) - x| = 2 \left| |I_n \cap [0, x]| - \frac{x}{2} \right| \leq \frac{1}{n}.$$

Hence

$$\sup_{x \in [0, 1]} |F_n(x) - x| \leq \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0.$$

But x is exactly the CDF of $Z \sim U(0, 1)$ on $[0, 1]$. Therefore $F_n(x) \rightarrow F_Z(x)$ at every continuity point of F_Z , and so $X_n \xrightarrow{d} Z$ by definition.

- b) (10 points) If $Z_N = \frac{\sum_{i=1}^N X_i - \mu N}{\sqrt{N}}$ is the normalized sum of an IID sequence with mean μ and variance σ^2 , and $Y \sim \mathcal{N}(0, \sigma^2)$ is an independent Gaussian variable, then $Z_N \xrightarrow{d} Y$.

- (a) False. $P(|Z_N - Y| \geq \varepsilon) \geq P(Z_N \geq \varepsilon, Y \leq 0) = P(Z_N \geq \varepsilon)P(Y \leq 0)$, since $Z_N \geq \varepsilon, Y \leq 0$ implies that $|Z_N - Y| \geq \varepsilon$. Then, when $N \rightarrow \infty$, $P(Z_N \geq \varepsilon)P(Y \leq 0)$ does not go to zero but to some positive number, using the CLT. Hence the definition for $Z_N \xrightarrow{P} Y$ does not hold.
- c) (10 points) Let $X_1, X_2, \dots$ be a sequence of random variables. If the sequence converges in distribution to a random variable X which is constant with probability 1 (i.e., $X = c$ for a deterministic constant c), then the sequence does not necessarily converge to c in probability.

- (a) False. $F_X(x) = 0$ for all $x < c$ and $F_X(x) = 1$ for all $x \geq c$. Since we have convergence in distribution, then

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq c - \varepsilon) = F_X(c - \varepsilon) = 0$$

and also

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(X_n \leq c + \frac{\varepsilon}{2}\right) = F_X\left(c + \frac{\varepsilon}{2}\right) = 1$$

and also

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \geq c + \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}\left(X_n > c + \frac{\varepsilon}{2}\right) = \lim_{n \rightarrow \infty} \left(1 - \mathbb{P}\left(X_n \leq c + \frac{\varepsilon}{2}\right)\right) = 0$$

note that $\mathbb{P}(X_n \geq c + \varepsilon) \neq 1 - \mathbb{P}(X_n \leq c + \varepsilon)$ but instead $\mathbb{P}(X_n \geq c + \varepsilon) = 1 - \mathbb{P}(X_n < c + \varepsilon)$, and $\mathbb{P}(X_n < c + \varepsilon)$ is not the CDF at point $c + \varepsilon$. Using the union bound

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - c| \geq \varepsilon) \leq \lim_{n \rightarrow \infty} (\mathbb{P}(X_n \geq c + \varepsilon) + \mathbb{P}(X_n \leq c - \varepsilon)) = 0$$

- d) (10 points) Let $X_1, X_2, \dots$ be a sequence of random variables. If $X_n \xrightarrow{d} X$ and $X_n \xrightarrow{d} Y$ (that is, the sequence converges in distribution both to X and to Y), then necessarily

$$\mathbb{P}(X = Y) = 1.$$

- (a) False. Let X_n be an IID sequence of fair coins (+1 or -1 with equal probability). Let X be a fair coin, and $Y = -X$. Then $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ but $\mathbb{P}(X = Y) = 0$.
- e) (10 points) Let X_n and Y_n be two sequences of random variables, and let X and Y be random variables. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$, then $X_n + Y_n \xrightarrow{d} X + Y$.
- (a) False. Let X_n and Y_n be IID sequence of fair coins (+1 or -1 with equal probability), independent of each other. Let X be a fair coin, and $Y = -X$. Then $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ but $X + Y = 0$ with probability one, while $X_n + Y_n = 0$ with probability $\frac{1}{2}$.
- f) (10 points) If a sequence of random variables X_n converges in probability to a random variable X , and all variables in the sequence (including the limit variable X) have finite variance, then it is necessarily true that the sequence of variances converges to the variance of the limit, i.e.:

$$\lim_{n \rightarrow \infty} \text{Var}(X_n) = \text{Var}(X).$$

- (a) False. Consider the sequence $X_n = \begin{cases} n, & \text{with probability } 1/n, \\ 0, & \text{with probability } 1 - 1/n. \end{cases}$. Then for $X = 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(X_n = n) = \frac{1}{n} \rightarrow 0,$$

so $X_n \rightarrow X$ in probability. However, $\mathbb{E}\{X_n\} = 1$ and $\mathbb{E}\{X_n^2\} = n$. Hence $\lim_{n \rightarrow \infty} \text{Var}(X_n) = \infty$ while $\text{Var}(X) = 0$.

g) (10 points) If $X_n \xrightarrow{p} X$, and Y has the same distribution as X , then $X_n \xrightarrow{d} Y$

(a) True. $X_n \xrightarrow{p} X$ implies that $X_n \xrightarrow{d} X$, and $X_n \xrightarrow{d} Y$ has the same definition if X and Y have the same distribution (i.e., the same CDF).

h) (10 points) Let $X_1, X_2, \dots$ be a sequence of IID Bernoulli random variables (coin flips taking values 0 or 1). We define a new sequence Y_N as their running average: $Y_N = \frac{1}{N} \sum_{i=1}^N X_i$. The sequence Y_N converges in L^2 .

(a) True. By definition

$$\mathbb{E} \left\{ \left(Y_N - \frac{1}{2} \right)^2 \right\} = \mathbb{E} \left\{ \left(\frac{1}{N} \sum_{i=1}^N \left(X_i - \frac{1}{2} \right) \right)^2 \right\} \stackrel{(a)}{=} \frac{1}{N^2} \mathbb{E} \left\{ \sum_{i=1}^N \left(X_i - \frac{1}{2} \right)^2 \right\} = \frac{1}{4N}$$

where (a) uses that

$$\mathbb{E} \left\{ \sum_{i=1}^N \left(X_i - \frac{1}{2} \right)^2 \right\} = \mathbb{E} \left\{ \sum_{i=1}^N \sum_{j=1}^N \left(X_i - \frac{1}{2} \right) \left(X_j - \frac{1}{2} \right) \right\} = \mathbb{E} \left\{ \sum_{i=1}^N \left(X_i - \frac{1}{2} \right)^2 \right\}$$

since $X_1 - \frac{1}{2}, \dots, X_N - \frac{1}{2}$ are independent with mean zero, so $\mathbb{E} \left\{ \left(X_i - \frac{1}{2} \right) \left(X_j - \frac{1}{2} \right) \right\} = 0$ for all $i \neq j$. Thus, $\mathbb{E} \left\{ \left(Y_N - \frac{1}{2} \right)^2 \right\} \rightarrow 0$ as $N \rightarrow \infty$ and Y_N converges to $\frac{1}{2}$ in L^2 .

i) (10 points) If for every pair of indices n, m the random variables X_n and X_m have exactly the same marginal distribution ($F_{X_n}(x) = F_{X_m}(x)$ for all x), then the random process $X_1, X_2, \dots$ is a Stationary Process.

(a) False. Let $X_1, X_2, \dots$ be a sequence of random variables such that X_1 is a coin (zero or one). For all $n \geq 1$, we have $X_{n+1} = X_n$ with probability $1 - \frac{1}{n}$ and otherwise $X_{n+1} = 1 - X_n$ with probability $\frac{1}{n}$. Then X_n is a coin for each n , having the same CDF. However, $\mathbb{E}\{X_1 X_2\} = 0$ while $\mathbb{E}\{X_{1000} X_{1001}\}$ is almost $\frac{1}{2}$.

j) (10 points) Let $X_1, X_2, \dots$ be a sequence of random variables. If $X_n \xrightarrow{p} X$ and $X_n \xrightarrow{p} Y$, then $P(X = Y) = 1$.

(a) True. For all n ,

$$\begin{aligned} P(|X - Y| \geq 2\varepsilon) &\leq P(|X_n - X| + |X_n - Y| \geq 2\varepsilon) \leq \\ &P\left(\{|X_n - X| \geq \varepsilon\} \cup \{|X_n - Y| \geq \varepsilon\}\right) \leq P(|X_n - X| \geq \varepsilon) + P(|X_n - Y| \geq \varepsilon) \end{aligned}$$

and taking the limit $n \rightarrow \infty$ shows that $P(|X - Y| \geq 2\varepsilon) = 0$. Since this holds for any $\varepsilon > 0$, we must have $P(|X - Y| > 0) = 0$, which is equivalent to $P(X = Y) = 1$.

k) (10 points) Let $X_1, X_2, \dots$ be a sequence of random variables. If $\mathbb{E}\{|X_n - X|\} \leq \frac{1}{n^2}$ for all n , then $X_n \xrightarrow{a.s.} X$.

(a) True. Let $\varepsilon > 0$. By Markov's inequality $P(|X_n - X| \geq \varepsilon) \leq \frac{\mathbb{E}\{|X_n - X|\}}{\varepsilon} \leq \frac{1}{\varepsilon n^2}$. Hence $\sum_{n=1}^{\infty} P(|X_n - X| \geq \varepsilon) < \infty$, and as we proved in class, it implies that $X_n \xrightarrow{a.s.} X$.

l) (10 points) Let $X_1, X_2, \dots$ be a sequence of random variables such that $|X_n| \leq C$ for all n with probability 1, for $C > 0$. If $X_n \xrightarrow{P} X$ then $X_n \xrightarrow{a.s.} X$.

(a) False. Let $U \sim U(0, 1)$, and define

$$X_1 = \mathbf{1}_{[0,1]}(U), \quad X_2 = \mathbf{1}_{[0, \frac{1}{2}]}(U), \quad X_3 = \mathbf{1}_{[\frac{1}{2}, 1]}(U),$$

$$X_4 = \mathbf{1}_{[0, \frac{1}{4}]}(U), \quad X_5 = \mathbf{1}_{[\frac{1}{4}, \frac{1}{2}]}(U), \quad X_6 = \mathbf{1}_{[\frac{1}{2}, \frac{3}{4}]}(U), \quad X_7 = \mathbf{1}_{[\frac{3}{4}, 1]}(U),$$

and so on. At stage m , the interval $[0, 1]$ is partitioned into 2^m dyadic intervals of length 2^{-m} , and the corresponding indicators are listed consecutively. Then $0 \leq X_n \leq 1$ for all n . Moreover, $\mathbb{P}(X_n = 1) = 2^{-m} \rightarrow 0$, where m is the stage containing X_n . Hence $X_n \xrightarrow{P} 0$. However, for every $u \in (0, 1)$, at each stage m there is exactly one dyadic interval containing u . Therefore $X_n(u) = 1$ infinitely often. Also $X_n(u) = 0$ infinitely often. Thus the sequence $\{X_n(u)\}$ does not converge. Hence $X_n \not\xrightarrow{a.s.} 0$.